

Practice questions

1. Differentiate between rule-based systems (traditional programming) and learning-based systems (machine learning) in AI with an example each.

Rule-based systems in AI use manually written “if-else” rules to make decisions and work well for simple, fixed problems. For example, a traditional spam filter might flag emails with phrases like “win money.” However, they cannot adapt to new or unseen data.

Learning-based systems, on the other hand, are a part of machine learning and learn patterns from data. For example, a spam filter trained on many labeled emails can detect spam based on learned patterns. These systems are more flexible and can adapt to complex, real-world data.

2. What is overfitting in machine learning? How does cross-validation (not included) help to avoid it?

Overfitting in machine learning occurs when a model learns the training data **too well**, including its **noise and random fluctuations**, rather than just the underlying patterns. As a result, the model performs very well on the training data but fails to generalize to unseen or new data, leading to poor performance on the test set. Overfitting is especially common when the model is too complex, the dataset is small, or the training continues for too long without proper regularization.

One common technique to **detect and reduce overfitting** is **cross-validation**. Although not a direct method to prevent overfitting, cross-validation helps in identifying it early by providing a more reliable estimate of the model's performance on unseen data. In k-fold cross-validation, the dataset is split into k parts or folds. The model is trained on $(k-1)$ folds and tested on the remaining one. This process is repeated k times, each time with a different test fold. The average performance across all folds gives a better indication of how the model will perform in real-world situations. By analyzing cross-validation results, we can detect if the model is overfitting and adjust accordingly by using simpler models, collecting more data, or applying regularization techniques.

Practice questions

3. List two differences between classification and clustering (not included) with real-world examples

Classification and **clustering** are both types of machine learning, but they differ in approach and purpose.

Classification is a **supervised learning** method, where the model is trained on labeled data (data with known output classes). The goal is to assign new data points to one of the predefined categories.

Example: Classifying emails as *spam* or *not spam* using a labeled dataset of past emails.

Clustering is an **unsupervised learning** method, where the data has no labels. The algorithm groups similar data points into clusters based on patterns or similarities without any predefined categories.

Example: Grouping customers into segments based on purchasing behavior for targeted marketing.

4. Define a "model" in machine learning. Explain how it is evaluated using accuracy and F1-score (harmonic mean of precision and recall - not included).

In **machine learning**, a **model** is a mathematical representation or program that is trained on data to recognize patterns and make predictions or decisions. It learns from input data and adjusts its internal parameters to minimize errors and improve accuracy over time.

Once trained, a model is evaluated to measure how well it performs on unseen data. One common metric is **accuracy**, which is the ratio of correctly predicted instances to the total number of predictions. It is simple and effective when the classes are balanced.

However, in cases of **imbalanced data**, accuracy alone can be misleading. In such cases, **F1-score** is used, which balances both **precision** (how many predicted positives are actually correct) and **recall** (how many actual positives were correctly predicted). The F1-score provides a more reliable evaluation when both false positives and false negatives matter.

Practice questions

5. A logistic regression model is trained to detect heart disease. The logit function is given by:

$$z = -1.5 + 0.7x$$

What is the probability of heart disease for a person with $x = 4$?

$$\hat{y} = p = 1 / (1 + e^{-z})$$

$$z = -1.5 + 0.7x = -1.5 + 0.7(4) = 1.3$$

$$p = 1 / (1 + e^{-z}) = 1 / (1 + e^{-1.3}) = 0.78$$

6. A linear regression model is defined as:

$$Y = 100 + 40X_1 + 25X_2$$

Predict Y if $X_1 = 8$ and $X_2 = 2$.

$$Y = 100 + 40(8) + 25(2) = 470$$

7. What is the purpose of the learning rate in gradient descent? What might happen if it's too large?

The **learning rate** in gradient descent is a hyperparameter that controls how **big the steps** are when updating the model's parameters (like weights) during training. It determines how quickly or slowly the model learns from the data.

If the **learning rate is too large**, the model may take steps that are too big and **overshoot the minimum** of the cost function. This can cause the model to **fail to converge**, leading to unstable or inaccurate results..

Choosing an appropriate learning rate is crucial for efficient and effective training of the model.

8. Find slope, intercept and MSE? You are given with following dataset:

Practice questions

Practice Question for Students

You are provided the following dataset:

x	y
4	2
7	4
11	7
18	12
22	14
28	18

Practice questions

Q8

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
4	2	-11	-7.5	82.5	121
7	4	-8	-5.5	44	64
11	7	-4	-2.5	10	16
18	12	3	2.5	7.5	9
22	14	7	4.5	31.5	49
28	18	13	8.5	110.5	169
				$\Sigma 286$	$\Sigma 428$

$$m = \frac{\Sigma (x - \bar{x})(y - \bar{y})}{\Sigma (x - \bar{x})^2}$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{90}{6} = 15$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{57}{6} = 9.5$$

$$m = \frac{286}{428} = 0.668$$

$$b = \bar{y} - m\bar{x}$$

$$= 9.5 - (0.668)(15)$$

$$b = -0.523$$

Practice questions

9. Find the updated values of θ_0 and θ_1 if you're given with initial values and derivatives of loss function with respect to θ_0 and θ_1 ?

You are performing gradient descent to minimize the mean squared error for a linear regression model:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

At a certain iteration, you are given the following:

- **Initial values:**

$$\theta_0 = 1.0, \quad \theta_1 = 2.0$$

- **Learning rate** $\alpha = 0.1$

- **Gradients** at this point:

$$\frac{\partial J}{\partial \theta_0} = 4.0, \quad \frac{\partial J}{\partial \theta_1} = -2.0$$

Task:

Using one step of gradient descent, compute the **updated values** of θ_0 and θ_1 .

$$\theta_0 \text{ (updated)} = \theta_0 \text{ (initial)} - \alpha * dJ/d\theta_0$$

Practice questions

Q9.

Initial Values

$$\theta_0 = 1.0$$

$$\theta_1 = 2.0$$

Learning Rate = $\alpha = 0.1$

Gradients at this point = $\frac{\partial J}{\partial \theta_0} = 4.0$, $\frac{\partial J}{\partial \theta_1} = -2.0$

Solution:-

Updated Values

$$\theta_0 = \theta_0 - \alpha \frac{\partial J}{\partial \theta_0}$$

$$= 1.0 - 0.1(4.0)$$

$$= 0.6$$

$$\theta_1 = \theta_1 - \alpha \frac{\partial J}{\partial \theta_1}$$

$$= 2.0 - 0.1(-2.0)$$

$$= 2.2$$